Natural Language Processing – NLP Project 1 (50%) — Cross-Cultural Knowledge Evaluation (BLEnD / SemEval-Style)

Module: Natural Language Processing
Weight: 50% of overall module mark
Submission Deadline: [10 April 2026]

1. Context and Motivation

Large Language Models (LLMs) often demonstrate significant gaps in culture-specific everyday knowledge, particularly for under-resourced regions and languages. Many existing benchmarks rely heavily on Western-centric data (e.g., Wikipedia), failing to capture everyday realities such as local foods, school traditions, work practices, or holidays.

This project is inspired by BLEnD (Benchmark for LLMs on Everyday Knowledge in Diverse Cultures and Languages). You will build and evaluate an NLP system that demonstrates:

- Technical proficiency in multilingual inference
- Careful evaluation and reproducibility
- Critical awareness of cultural bias and responsible AI

2. Project Structure

You must complete **one track in depth**.
Completing both tracks may be considered for distinction-level performance.

Track A — Short Answer Questions (SAQ)

Input: Question + locale (e.g., `ga-IE`, `ar-DZ`, `zh-CN`)
Output: Short answer in the same local language

Evaluation:
An answer is correct if it matches *any* human-annotated reference after appropriate normalisation.

You must:

- Implement a robust answer normalisation strategy (Unicode normalisation, token normalisation, stemming/lemmatisation if appropriate, or embedding-based similarity).
- Justify your chosen matching method.

Track B — Multiple Choice Questions (MCQ)

Input: Question (English) + target locale + 4 culturally distinct options
Output: Selected option (A/B/C/D or 0–3)

Evaluation:
Accuracy per locale and overall.

You must:

- Enforce structured output (model must return only the selected option).
- Analyse performance differences across locales.

3. Minimum Requirements

You must evaluate on **at least four locales**, including:

- At least **one non-Latin script locale** (e.g., Arabic, Chinese, Japanese, Thai)
- At least **one under-represented or low-resource locale**
- Two additional locales of your choice

4. System Requirements

You must implement a **local or cloud-based inference pipeline** using open-source/open-weight models.

Model Constraints

- Commercial APIs (e.g., GPT-4, Claude) are **not permitted**.
- Recommended model size: **7B–13B class**.
- Larger models may be used but are not required.
- All decoding must use **temperature = 0** for reproducibility.

5. Required Project Components

A. Baseline (Required)

Implement a clear baseline system.
Examples:

- Direct prompting baseline
- Simple multilingual transformer inference
- Basic retrieval-augmented approach
- Structured constrained decoding

B. Improvements Beyond Baseline (Required)

You must implement **at least two meaningful improvements**, such as:

- Persona-based prompting
- Structured JSON outputs
- Retrieval augmentation (RAG)
- Confidence estimation or abstention mechanism
- Locale-aware prompting
- Calibration strategy
- Error-driven prompt refinement

You must demonstrate whether these improvements help (or explain why not).

C. Evaluation & Analysis (Required)

Your evaluation must include:

- Overall performance
- Performance by locale (table required)
- Gap vs best-performing locale
- At least **10 analysed failure examples**
- At least one case where the model produces a fluent but culturally incorrect answer

6. Responsible AI Section (Required)

Your report must include a dedicated section discussing:

- Cultural bias and uneven performance
- Risks of deployment in under-resourced contexts
- Overconfidence and hallucination risks
- Trust calibration (uncertainty, abstention, guardrails)
- Transparency: what users should be told

Concrete mitigation strategies must be proposed.

7. Reproducibility Requirements

You must ensure:

- Fixed random seeds
- Temperature = 0 decoding
- Clear model version documentation
- Exact locale list documented
- Clear data preprocessing steps

Work must be reproducible from your README instructions.

8. Deliverables

Submit a single ZIP file containing:

1) Report (PDF, 6–10 pages)

Must include:

1. Introduction & task description
2. Dataset and locale selection
3. Methodology
    - Baseline
    - Improvements
4. Evaluation
    - Tables (overall + per-locale)
5. Error analysis (≥10 examples)
6. Responsible AI reflection
7. Limitations & future work

Formatting:

- LREC-style formatting recommended (https://lrec2026.info/authors-kit/)
- LaTeX encouraged but not mandatory (clear academic formatting required)

2) Code

- Clean, readable Python code
- Logical structure (functions, modularity)
- `requirements.txt` or `environment.yml`

- `README.md` with step-by-step reproduction instructions

3) Prediction Files

- `predictions_saq.csv` (if applicable)
- `predictions_mcq.csv` (if applicable)

9. Marking Rubric (50% of the NLP Module)

Component	Weight
Reproducibility & Engineering Quality	20%
Baseline Implementation	10%
Improvements Beyond Baseline	20%
Evaluation & Per-Locale Analysis	25%
Error Analysis Depth	15%
Responsible AI Reflection	10%

Distinction-Level Indicators

- Completing both tracks
- Confidence calibration or abstention modelling
- Statistical comparison across locales
- Clear identification of systemic cultural bias patterns
- Particularly strong Responsible AI reasoning

10. Academic Integrity

- All code must be your own.
- Any AI assistance must be disclosed.
- Reports will be checked for plagiarism.

11. Late Submission Policy

Standard MTU late submission policy applies:

- Up to 2 weeks late with 10% penalty.
- Medical certificates must be submitted to the department secretary if applicable.

12. Final Submission Checklist

Before uploading to Canvas:

☐ Report PDF included
☐ Code included
☐ requirements.txt included
☐ README included
☐ Predictions file included
☐ Locales clearly listed
☐ Responsible AI section included